

CLASS: X

MAX.MARKS:80

SUBJECT: MATHEMATICS BASIC (241)

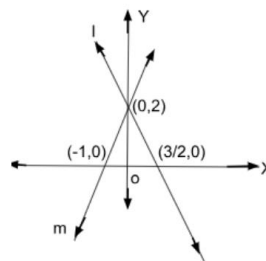
TIME: 3 HOURS

General Instructions:

1. This question paper contains 38 questions.
2. This Question Paper is divided into 5 Sections A, B, C, D and E.
3. In Section A, Questions no. 1-18 are multiple choice questions (MCQs) and questions no. 19 and 20 are Assertion- Reason based questions of 1 mark each.
4. In Section B, Questions no. 21-25 are very short answer (VSA) type questions, carrying 02 marks each.
5. In Section C, Questions no. 26-31 are short answer (SA) type questions, carrying 03 marks each.
6. In Section D, Questions no. 32-35 are long answer (LA) type questions, carrying 05 marks each.
7. In Section E, Questions no. 36-38 are case study based questions carrying 4 marks each with sub parts of the values of 1, 1 and 2 marks each respectively.
8. All Questions are compulsory. However, an internal choice in 2 Question of Section B, 2 Questions of Section C and 2 Questions of Section D has been provided.
9. Draw neat and clean figures wherever required.
10. Take $\pi = 22/7$ wherever required if not stated.
11. Use of calculators is not allowed.

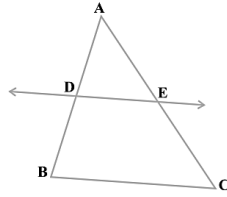
Section A

1. HCF of $(3^2 \times 5^2 \times 2)$, $(3^2 \times 5^3 \times 2^2)$ and $(3^4 \times 5 \times 2^3)$ is
a) 450 b) 90 c) 180 d) 630
2. The quadratic polynomial whose zeros are 2 and -3 is
a) $x^2 - x - 6$ b) $x^2 - 5x - 6$ c) $x^2 + x - 6$ d) $x^2 - 5x + 6$
3. The system of linear equations represented by the lines l and m is

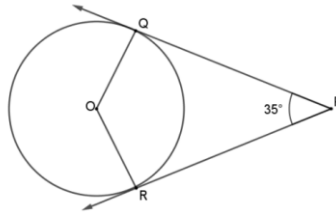


- a) Consistent with unique solution b) inconsistent
 - c) Consistent with three solutions d) consistent with many solutions
4. The discriminant of the quadratic equation $2x^2 - 3x - 5 = 0$ is
a) 43 b) -40 c) 7 d) 49
 5. The roots of the quadratic equation $7x^2 + 2x + 1 = 0$ are
a) Not real b) real and equal c) real and unequal d) irrational
 6. Two dice are rolled simultaneously. The probability of getting a doublet is
a) $\frac{1}{3}$ b) $\frac{1}{4}$ c) $\frac{1}{6}$ d) $\frac{1}{2}$
 7. The distance between the points $(3, 4)$ and the origin is
a) 7 b) 1 c) 25 d) 5
 8. The triangles which are always similar are
a) Equilateral b) isosceles c) right d) right isosceles

9. Given DE is parallel to BC in the figure given below and AD = 3 cm , DE = 5 cm and EC = 2 cm . Then AE = _____ cm



- a) 3.3 b) 1.2 c) 4 d) 1
10. The coordinates of the centre of a circle if the coordinates of the endpoints of one of its diameter are (4, 7) and (- 2 , 1) is
- a) (-2 , 8) b) (-4 , 16) c) (1 , 4) d) (4,1)
11. In the given figure, PQ and PR are tangents to a circle centred at O. If $\angle QPR = 35^\circ$ then $\angle QOR$ is equal to



- a) 70° b) 90° c) 135° d) 145°
12. If θ Lies between 0° and 90° and $\sin \theta = \cos \theta$ then $\theta =$ _____
- a) 30° b) 45° c) 0° d) 60°
13. $\sqrt{(1 + \cos \theta)(1 - \cos \theta)} =$ _____
- a) $\sin \theta$ b) $\cos \theta$ c) $\sec \theta$ d) $\operatorname{cosec} \theta$
14. The area of a quadrant with radius 7 cm is
- a) 77 sq.cm b) 154 sq.cm c) 38.5 sq.cm d) 48.5 sq.cm
15. If the slant height of a cone is 13 cm and its radius is 5 cm then the CSA of the cone will be _____sq.cm (Take $\pi = 3.14$)
- a) 65 b) 240 c) 188.4 d) 204.1
16. if the volume of a cone is equal to volume up for hemisphere given that the radii of both are the same then the relation between the height of the cone(h) and the radius(r) of the hemisphere is
- a) $h = 2r$ b) $h = r$ c) $r = 2h$ d) $h = 3r$
17. The Modal class of the given data is

Class	65-85	85-105	105-125	125-145	145-165	165-185	185-205
Frequency	4	5	13	20	14	7	4

- a) 105-125 b) 125-145 c) 145- 165 d) 185-205
18. A bag contains 5 blue , 6 green and 4 red balls. A ball is drawn at random. The probability of picking a blue ball is
- a) $\frac{1}{3}$ b) $\frac{2}{5}$ c) $\frac{4}{15}$ d) $\frac{2}{3}$

- The following are the options for the Assertion and Reasoning questions
 - Both assertion and reasoning are correct and the reason is the correct explanation for the assertion
 - both assertion and reasoning are correct and the reason is not the correct explanation for the assertion
 - the assertion is true but the reason is false
 - the assertion is false but the reason is true

19. Assertion (A): The point $(-1, 6)$ divides the line segment joining the points $(-3, 10)$ and $(6, -8)$ in the ratio $2 : 7$ internally.

Reason (R): Given three points, i.e. A, B, C form an equilateral triangle, then $AB = BC = AC$.

20. Assertion: Construction of cumulative frequency table is helpful in finding the median

Reason: The median class is determined by finding the class interval whose cumulative frequency is just greater than half the total of frequencies of the given grouped data

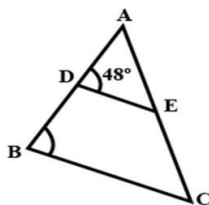
SECTION B

21. Find the LCM(91,117) ; given that $HCF(91,117) = 13$.

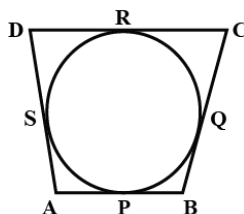
Or

Prove that $2 + 5\sqrt{3}$ is irrational. Given that $\sqrt{3}$ is irrational.

22. In the figure given below, if $AD = 6$ cm, $DB = 9$ cm, $AE = 8$ cm and $EC = 12$ cm and $\angle ADE = 48^\circ$. Find $\angle ABC$.



23. A circle touches all the four sides of a quadrilateral ABCD. Prove that $AB + CD = AD + BC$.



OR

The length of a tangent from a point A at 5 cm from the centre of the circle is 4 cm. Find the radius of the circle.

24. If $5 \sin \theta = 4$ then find the values of $\cos \theta$ and $\cot \theta$.

25. From a well shuffled deck of 52 cards, a card is chosen at random. Find the probability of getting

- a) an Ace card
- b) a Diamond

Or

A card is drawn at random from a well shuffled pack of 52 playing cards. Find the probability of getting neither a red card nor a queen.

SECTION C

26. Prove that $\sqrt{3}$ is irrational.

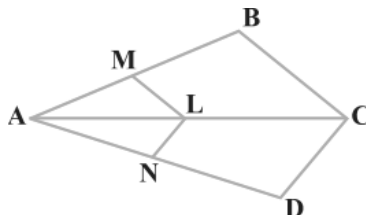
27. Verify the relationship between the zeroes and the coefficients of the quadratic polynomial $3x^2 + 10x + 7$.

28. Solve $4x^2 - 8x + 3 = 0$ using quadratic formula.

29. Prove the theorem: If a line is drawn parallel to one side of a triangle to intersect the other two sides in distinct points, the other two sides are divided in the same ratio.

Or

In the figure given below, $LM \parallel CB$ and $LN \parallel CD$. Prove that $\frac{AM}{AB} = \frac{AN}{AD}$.



30. Evaluate:

$$\frac{\cos 45^\circ}{\sec 30^\circ + \operatorname{cosec} 30^\circ}$$

31. In a circle of radius 21 cm, an arc subtends an angle of 60° at the centre. Find:

- (i) the length of the arc
- (ii) area of the sector formed by the arc

Or

The length of the minute hand of a clock is 14 cm. Find the area swept by the minute hand in 5 minutes.

SECTION D

32. Solve the following pair of linear equations for x and y :

$$3x + 5y = 13 \quad \text{and} \quad 4x - 3y = -2$$

Now, find the value of " p " if $5y = x + 3p$

OR

The taxi charges in a city consist of a fixed charge together with the charge for the distance covered. For a distance of 10 km, the charge paid is Rs.105 and for a journey of 15 km, the charge paid is Rs.155. What are the fixed charges and the charge per km?

33. From the top of a 7 m high building, the angle of elevation of the top of a cable tower is 60° and the angle of depression of its foot is 45° . Determine the height of the tower.

34. Prove that the radius of a circle is perpendicular to the tangent at the point of contact. Using the above theorem prove the following:

The angle between two tangents from an external point to a circle is supplementary to the angle subtended by the line segments joining the points of contact at the centre.

35. Thirty women were examined in a hospital by a doctor and the number of heartbeats per minute were recorded and summarised as follows. Find the mean heartbeats per minute for these women, choosing a suitable method. Also find the mode of the data.

Number of heartbeats per minute	65-68	68-71	71-74	74-77	77-80	80-83	83-86
Number of women	2	4	3	8	7	4	2

OR

The median of the following data is 525. Find the values of x and y , if the total frequency is 100.

Class interval	Frequency
0–100	2
100–200	5
200–300	x
300–400	12
400–500	17
500–600	20
600–700	y
700–800	9
800–900	7
900–1000	4

SECTION E

36. Rinku was very happy to receive a fancy jumbo pencil from his best friend Rohan on his birthday. Pencil is a basic writing tool, when sharpened its shape is a combination of cylinder & cone as given in the picture. Cylindrical pencil with conical head is a common shape worldwide since ages. Commonly pencils are made up of wood & plastic but we should promote pencils made up of eco-friendly material (many options available in the market these days) to save environment.



The dimensions of Rinku's pencil are given as follows: Length of cylindrical portion is 21cm. Diameter of the base is 1 cm and height of the conical portion is 1.2 cm Based on the above information, answer the following questions:

- Find the slant height of the sharpened part.(1 mk)
- Find curved surface area of sharpened part (in terms of π). (1mk)
- Find the total surface area of the pencil (in terms of π). (2 mks)

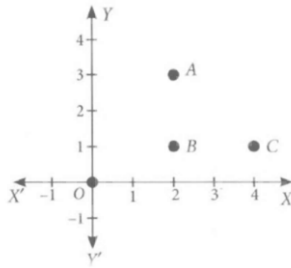
37. In a school, students thought of planting trees in and around the school to reduce air pollution. It was decided that the number of trees, that each section of each class will plant, will be the same as the class, in which they are studying, e.g., a section of Class I will plant 1 tree, a section of Class II will plant 2 trees and so on till Class XII. There are three sections of each class.



- Write the sequence formed showing the total number of trees planted by each class.
- Is it an arithmetic progression?
- If yes, find the first term and the common difference.
- How many trees will be planted by the students altogether?

38. Alia and Shagun are friends living on the same street in Patel Nagar. Shagun's house is at the intersection of one street with another street on which there is a library. They both study in the same school and that is not far from Shagun's house. Suppose the school is

situated at the point O, i.e., the origin, Alia's house is at A. Shagun's house is at B and library is at C. Based on the above information, answer the following questions.



- a) How far is Alia's house from Shagun's house? (1 mk)
- b) How far is the library from Shagun's house? (1 mk)
- c) How far is Alia's house from the library? (2mk)

or

Show that Alia's house, Shagun's house and library form an isosceles right triangle.
(2 mks)